Supplementary Material

Latent-variable reanalysis of cognitive group differences in Grassi et al. (2025)

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1 Overview

This supplement documents the latent-variable reanalysis accompanying the commentary on *Do Musicians Have Better Short-Term Memory Than Nonmusicians? A Multilab Study* (Grassi et al., 2025). The goal of the reanalysis is not to dispute the original empirical pattern. Rather, it is to ask at what level that pattern is most meaningfully interpreted.

Our guiding claim is simple. Once the cognitive measures are treated as fallible indicators of a shared cognitive dimension, the musician versus nonmusician contrast is most parsimoniously described as a difference on a general latent factor, `g`, plus a limited amount of remaining task-specific variance, most clearly for Melody span, and more modestly for vocabulary in the multilevel specification.

To keep this supplement readable, we report only the information that is substantively necessary for evaluating that claim. Full raw console output is intentionally omitted from the main document. The tables in the `/Tables` subfolder and the figures in the `/Figures` subfolder are used directly below.

2 Data and preliminary checks

The reanalysis used the same matched sample reported in the target article, namely 600 musicians and 600 nonmusicians. Before moving to latent models, it is useful to document the matching variables and the main covariate included in the SEMs.

Table 1: Table S1. Preliminary group comparison for matching variables and family SES.

Measure	Musicians	Nonmusicians	Cohen's d	95% CI
Age (years)	22.11	22.15	-0.01	[-0.13, 0.10]
Years of education	15.88	15.83	0.02	[-0.09, 0.14]
Family SES	50.06	44.90	0.34	[0.22, 0.45]

Age and years of education were, as intended, essentially matched across groups. By contrast, family SES was somewhat higher among musicians, with a small standardized difference, $d = 0.34$. Accordingly, SES was carried into the latent models as a covariate rather than treated as ignorable.

3 Why a latent-variable reanalysis is warranted

The commentary argues that interpretation should start from the well-known tendency of cognitive measures to correlate positively. In the present dataset this premise is straightforwardly supported. Raven, vocabulary, n-back, digit span, spatial span, and melody span all show mostly positive associations, and SES is also positively related to the cognitive indicators. In other words, the data exhibit exactly the kind of positive manifold that motivates a common-factor representation.

Table 2: Table S2. Exported cross-group correlation matrix for the standardized cognitive indicators and SES. The upper and lower triangles correspond to the two groups as saved by the preprocessing script.

	waisvocabulary	raven	nback	digit_span	spatial_span	melody_span	ses_nuclear_family
waisvocabulary		0.27	0.23	0.12	0.03	0.05	0.10
raven	0.24		0.34	0.20	0.34	0.16	0.12
nback	0.14	0.28		0.20	0.19	0.11	0.06
digit_span	0.12	0.23	0.23		0.31	0.24	0.09
spatial_span	-0.01	0.24	0.19	0.29		0.20	0.05
melody_span	0.13	0.11	0.14	0.41	0.12		0.09
ses_nuclear_family	0.08	0.18	0.13	0.16	0.04	0.06	

The correlations are not identical across groups, but they are similar enough in overall pattern to justify formal invariance testing. For the purposes of the commentary, the key implication is more basic: a task-by-task mean comparison is not the most informative starting point, because part of any observed task difference is likely to reflect shared variance across tasks.

4 Modeling strategy

Two complementary SEM strategies were used.

First, we fitted a **multigroup CFA / invariance sequence**. The latent factor `g` was defined by Raven, Wechsler Vocabulary, n-back, Spatial span, Digit span, and Melody span. The latent variance of `g` was fixed to 1 in both groups. SES was specified as a predictor of `g`. Three residual covariances were included throughout: Vocabulary with Spatial span, Spatial span with Digit span, and Digit span with Melody span.

Second, because the original study was multilab and cross-national, we fitted a **multilevel SEM** with clustering by country. This model cannot reproduce the exact multigroup design simultaneously, but it provides a useful converging test. The crucial question here was whether the group effect is best placed only on the observed variables, only on `g`, or on `g` plus a very small number of targeted residual paths.

5 Multigroup CFA results

5.1 Model comparison

Table 3: Table S3. Fit indices for the multigroup CFA sequence.

Model	Description	ChiSquare	df	p	RMSEA	CFI	BIC	Delta_BIC
Configural	Same factor structure across groups	31.913	22	0.079	0.027	0.986	19525.72	53.387
Metric	Equal loadings	42.326	28	0.040	0.029	0.979	19493.60	21.259
Scalar	Equal intercepts	292.244	33	0.000	0.114	0.626	19708.07	235.728
Partial scalar	Equal intercepts except Melody span	55.698	32	0.006	0.035	0.966	19478.61	6.272
Partial scalar + equal SES -> g	Equal loadings and intercepts except Melody span, with equal SES -> g regression	56.844	33	0.006	0.035	0.966	19472.67	0.328
Partial scalar + equal residual covariances	ScalarPartial_1 plus equal residual covariances	77.787	36	0.000	0.044	0.940	19472.34	0.000
Strict	Equal residual variances	128.276	39	0.000	0.062	0.871	19501.56	29.219
Equal latent means	Equal latent means	100.871	34	0.000	0.057	0.904	19509.60	37.264

The multigroup results are structurally clear.

Configural and metric invariance were both acceptable, which indicates that the one-factor measurement structure is broadly similar across musicians and nonmusicians. Full scalar invariance, however, fit very poorly. This matters, because it shows that the group contrast cannot be reduced to a simple latent mean difference under fully equal intercepts.

Once the Melody-span intercept was freed, fit improved dramatically and returned to an acceptable range. Adding the equality constraint on the SES regression did not materially damage fit, and this model remained slightly cleaner than the still more constrained residual-covariance model. Although the latter had the numerically smallest BIC, the difference was trivial (ΔBIC = 0.33), whereas approximate fit was somewhat worse. For interpretation, therefore, the more parsimonious **partial-scalar model with equal SES -> g regression** was retained.

Finally, when latent means were constrained equal, fit worsened again. This is the central substantive result of the multigroup analysis: after accounting for measurement structure and partial intercept noninvariance, a group difference remains at the latent-factor level.

5.2 Preferred multigroup representation

Figure S1 summarizes the final substantive picture. The group contrast is represented in two places. First, musicians are higher on the latent factor g. Second, Melody span shows a remaining indicator-specific intercept difference that is not absorbed by g.

Table 4: Table S4. Key estimates from the preferred multigroup model.

Parameter	Estimate	Interpretation
Latent mean difference in g (musicians - nonmusicians)	0.514	About half a latent SD

Parameter	Estimate	Interpretation
Residual mean difference in Melody span intercept	0.851	Additional Melody-span advantage beyond g, in the standardized observed metric
Standardized SES -> g (nonmusicians)	0.220	Small positive SES association with g
Standardized SES -> g (musicians)	0.198	Small positive SES association with g

The latent mean difference was approximately 0.51 latent SD units, which is not a task-specific effect but a general-factor effect. Because the latent variance of `g` was fixed to 1 in both groups, this contrast can be read directly as a difference of about half a latent standard deviation. At the same time, Melody span retained an additional intercept shift of roughly 0.85 units in the standardized observed metric. Thus, the musical-memory task is not simply another interchangeable indicator of the same group contrast, it contains an obvious residual advantage beyond the general cognitive factor.

The SES effect on `g` was small and positive in both groups, and the standardized coefficients were very similar. This supports the decision to retain a common SES regression in the preferred multigroup model.

5.3 Interpretation of the multigroup findings

The multigroup sequence supports a layered interpretation.

The broad musician advantage in the cognitive battery is largely captured by a single latent dimension. That is the main result. However, the failure of full scalar invariance also shows that not every group difference should be read as a pure reflection of `g`. Melody span carries additional group-related variance, which is exactly what one would expect from a task that is closest to musical expertise.

This pattern is theoretically preferable to a flat task-by-task reading. Instead of saying that musicians have separate advantages in several partially disconnected memory tasks, the latent model suggests a more economical account: a general cognitive difference plus one especially salient music-specific residual deviation.

6 Multilevel SEM results

6.1 Model comparison

Table 5: Table S5. Fit indices for the multilevel SEM sequence, with clustering by country.

Model	Description	ChiSquared	df	p	RMSEA	CFI	BIC	Delta_BIC
Latent g only	group differences and SES modeled only at the latent g level	297.055	31	0.000	0.085	0.777	19362.10	228.550
Observed variables only	group differences and SES modeled only at observed-variable level	34.588	21	0.031	0.023	0.989	19170.53	36.984

Model	Description	ChiSquare	df	p	RMSEA	CFI	BIC	Delta_BIC
Latent g + broad residual paths	group differences and SES modeled at latent g plus observed-variable level (except digit span)	34.588	21	0.031	0.023	0.989	19170.53	36.984
Latent g + targeted residual paths	group differences and SES modeled at latent g plus group differences direct effects on vocabulary and melody	54.324	29	0.003	0.027	0.979	19133.55	0.000

The multilevel models serve as a robustness check under a design that partly respects the clustered structure of the original dataset. The results again point to a mixed but clearly ordered solution.

A model placing the group effect only on g fit poorly. A model placing all group differences only at the observed-variable level fit much better. However, the best BIC was obtained by the targeted hybrid model, which combined a latent group effect on g with direct residual paths only to Wechsler Vocabulary and Melody span. This is exactly the pattern one would expect if the main contrast is general but not perfectly exhaustive.

6.2 Preferred multilevel representation

Table 6: Table S6. Key standardized effects from the preferred multilevel model. Absolute values are reported for ease of interpretation, because the sign depends on the chosen coding of the group variable.

Parameter	Estimate	Interpretation
Latent group effect on g	0.418	Latent advantage favoring musicians
Residual group effect on Wechsler Vocabulary	0.082	Small residual task-specific difference
Residual group effect on Melody span	0.430	Moderate residual task-specific difference
Standardized SES -> g	0.227	Small positive SES association with g

In the preferred multilevel model, the group effect on g was substantial, about 0.42 SD in standardized terms. Beyond that latent effect, there was only a small residual path to vocabulary and a larger residual path to Melody span. SES again predicted g positively.

This result is useful because it converges with the multigroup CFA while not being identical to it. The multilevel model still says that the main group contrast sits at the common-factor level. Yet it also suggests that a small amount of indicator-specific variance remains, concentrated especially in Melody span and, to a lesser extent, vocabulary.

7 Integrative interpretation

Taken together, the two reanalyses support the same broad conclusion.

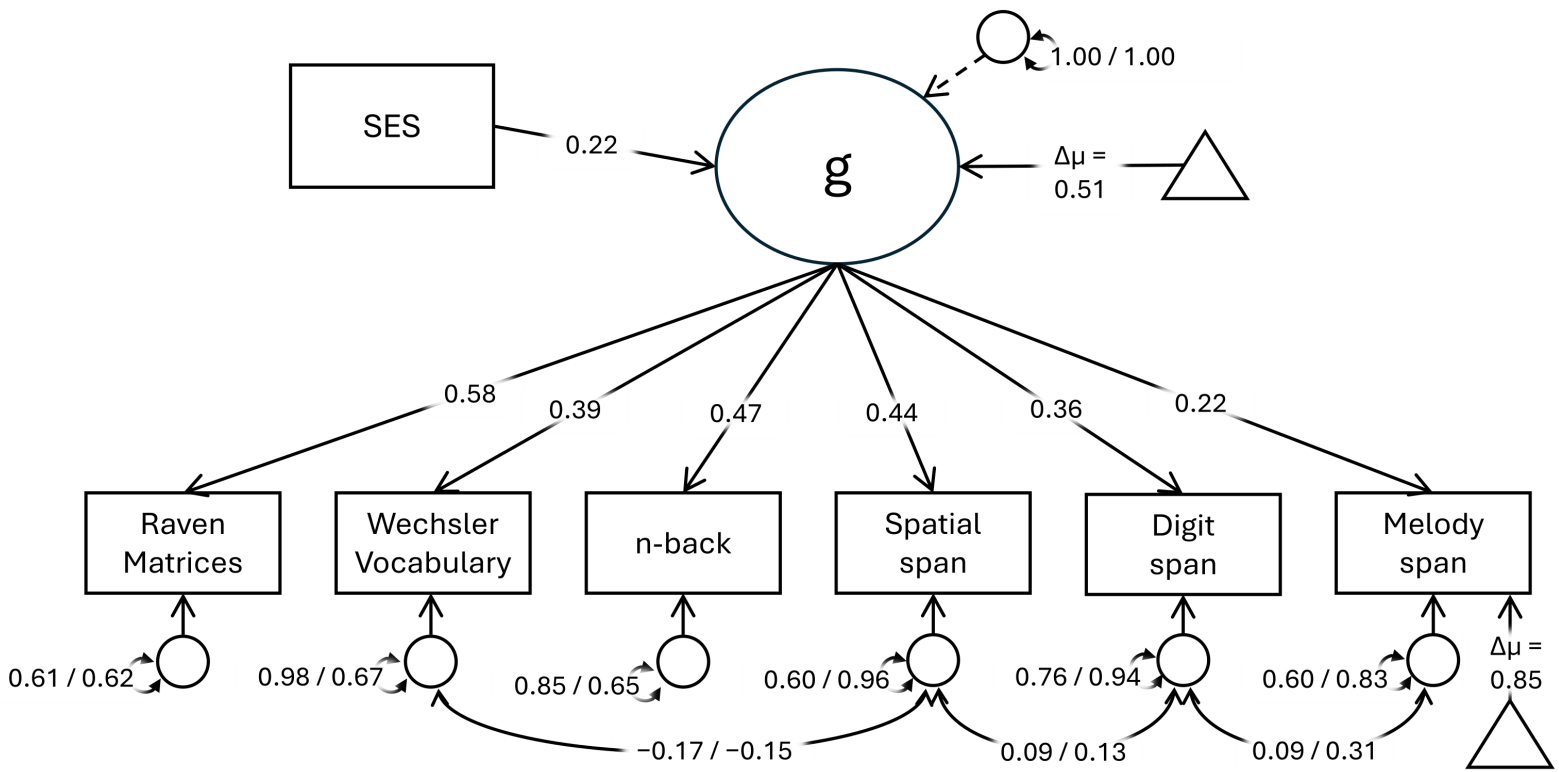

Figure 1: Figure S1. Preferred multigroup representation. Dual values separated by a slash denote the two groups for group-specific quantities such as residual variances or residual covariances.

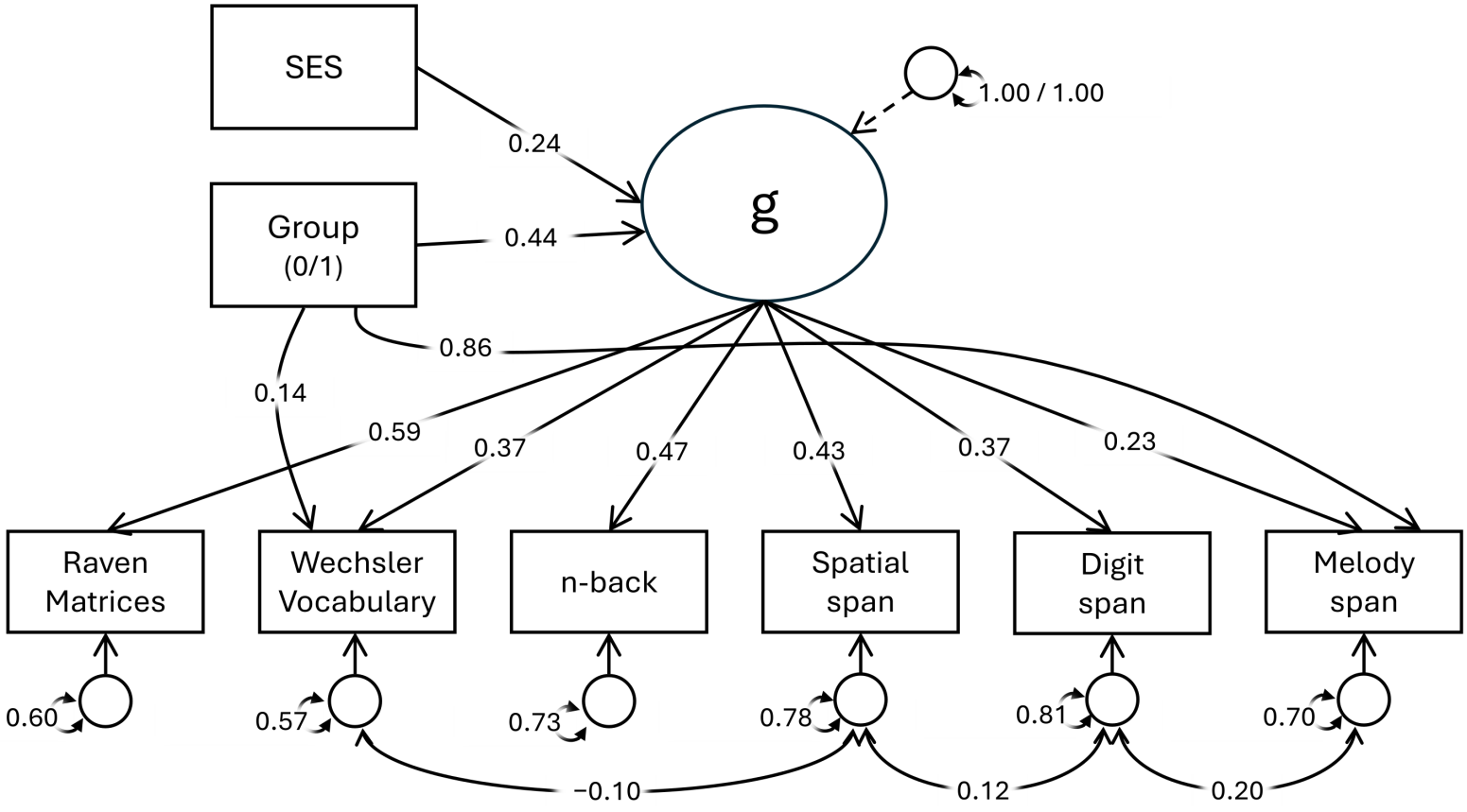

Figure 2: Figure S2. Preferred multilevel SEM representation. The sign of the raw group coefficients depends on the coding of the group variable; the diagram displays the effects in the direction favoring musicians.

The musician versus nonmusician contrast in this battery is not best described as a collection of separate advantages in verbal, visuospatial, executive, and musical short-term memory tasks. A more coherent and parsimonious reading is that musicians differ primarily on a **general cognitive factor** defined by the shared variance among the tasks. Once this common factor is modeled, the only clearly substantial residual deviation is the expected one, namely Melody span. A smaller vocabulary-specific deviation appears in the multilevel model, but it is much weaker than the latent effect and clearly secondary to it.

This matters for interpretation. The original empirical facts remain intact, but their organization changes. The results do not force a fragmented task-by-task narrative. They are better summarized as evidence for a broad latent difference, with task-specific excess concentrated where theory would most expect it, namely in the music-relevant indicator.

8 Reproducibility note

This `.Rmd` file is intentionally concise. It uses the precomputed tables stored in `/Tables` and the diagrams stored in `/Figures`, rather than re-running the full SEM workflow during knitting. This keeps the supplementary document easy to audit and stable across machines.

The underlying analyses were based on the same dataset used in the commentary project and were originally carried out in `lavaan`. Full scripts, raw console outputs, and intermediate model objects can be archived separately if desired. The present supplement is designed to be the readable technical companion to the commentary, not the full analysis log.

9 Session information

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 22631)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United Kingdom.utf8
## [2] LC_CTYPE=English_United Kingdom.utf8
## [3] LC_MONETARY=English_United Kingdom.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United Kingdom.utf8
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
```

```
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] knitr_1.50
##
## loaded via a namespace (and not attached):
##  [1] compiler_4.5.1    fastmap_1.2.0     cli_3.6.5         tools_4.5.1
##  [5] htmltools_0.5.8.1 rstudioapi_0.17.1 yaml_2.3.10       tinytex_0.56
##  [9] rmarkdown_2.30    xfun_0.51         digest_0.6.37     rlang_1.1.7
## [13] evaluate_1.0.3
```